Miguel Méndez Sandín, Ph.D.

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Born	1990/09/07, in Gijón, Asturias, Spain
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Other websites	Google Scholar: Miguel Mendez Sandin GitHub: MiguelMSandin Lead coordinator of "The Life of Retaria Seminar Series": thelifeofretaria.github.io/

Professional experience

2024/09/01 - 2026/08/31	**Marie Sklodowska-Curie Actions Postdoctoral Fellow**, Institute of Evolutionary Biology, CSIC - UPF, Spain *Hosts*: Daniel J. Richter and Ramon Massana *Project*: The hidden majority: high-resolution imaging and genomics of the most ecologically important microbial eukaryotes in the ocean
2023/03/01 - 2024/08/31	***Beatriu de Pinós* Postdoctoral Fellow**, Institute of Evolutionary Biology, CSIC - UPF, Spain *Hosts*: Daniel J. Richter and Ramon Massana *Project*: The hidden majority: high-resolution imaging of the most ecologically important microbial eukaryotes in the ocean
2022/09/08 - 2023/02/28	**Parental leave**
2020/09/15 - 2022/09/07	**Postdoctoral researcher**, Dept. of Organismal Biology, Uppsala University, Sweden *Advisor*: Fabien Burki *Project*: Eco-evolution of microbial eukaryotes, insights from metabarcoding surveys *Experience*: Diversification analysis, networks analysis, large data mining, lecturer, workshop organisation, fluorescence in situ hybridization
2016/10/01 - 2019/11/30	**Ph.D. Student**, Biological Station of Roscoff, CNRS - Sorbonne University, France *Supervisor*: Fabrice Not *Project*: Diversity and Evolution of Nassellaria and Spumellaria (Radiolaria) *Experience*: Single-cell molecular approaches, phylogenetics, molecular clock, metabarcoding and next generation sequencing analysis, scanning electron and confocal microscopy, student supervision
2016/03/01 - 05/31	**Research assistant**, Biological Station of Roscoff - Sorbonne University, France *Supervisor*: Sophie Martin *Project*: Impact of global and local changes in Maerl, red algae *Experience*: Experimental development
2015/10/05 - 2016/01/29	**Research assistant**, Oviedo University, Spain *Supervisor*: Eva Garcia-Vazquez *Project*: Top-down regulation in freshwater communities suggested from environmental DNA. A case study in mountain stream communities *Experience*: Field work, environmental DNA sampling and analysis
2015/01/12 - 06/30	**Master thesis**, Oviedo University, Spain *Supervisor*: Consolación Fernández *Project*: Long-term changes in low intertidal rocky shore assemblages in the coast of Cape *Peñas* (North of Spain). *Experience*: Multivariate analysis

Education

2016/10/01 - 2019/11/31	**Ph.D. in molecular diversity and evolution**, Biological Station of Roscoff - Sorbonne University, France. *Supervisor*: Fabrice Not. Awarded on **November 22, 2019**
- 2019/01/28 - 02/09	Visiting scientist: Institute of Marine Science, CSIC, Barcelona (Spain)
- 2018/10/15 - 12/07	Visiting scientist: National Institute of Water and Atmospheric Research, Wellington (New Zealand)
- 2017/10/30 - 11/03	Visiting scientist: Tropical Biosphere Research Center, Okinawa (Japan)
- 2016/10/05 - 10/28	Visiting scientist: Oceanographic Observatory of Villefranche sur Mer, CNRS (France)
2013/09/02 - 2015/06/30	**International MSc in Marine Biodiversity and Conservation** (EMBC+)
- 2015/01/12 - 06/30	Oviedo University, Spain. Master Thesis. *Supervisor*: Consolación Fernández
- 2014/09/01 - 12/19	Sorbonne University, France
- 2014/04/07 - 04/18	Gothenburg University, Sweden. Spring School. *Supervisor*: Rick Officer
- 2013/09/02 - 2014/06/27	Algarve University, Portugal
2008/09/29 - 2013/06/04	**BSc in Biology**, Ecology speciality. 311 ECTS. Oviedo University, Spain
- 2013/01/14 - 07/31	Student intern: Ecology department. *Supervisor*: Jose Luis Acuña
- 2012/09/03 - 12/21	ERASMUS: Klaipeda University, Lithuania
2006/09/04 - 2008/06/31	**International Baccalaureate**, IES Real Instituto de Jovellanos, Gijón, Spain

Fellowships and grants

2024	CGUE 2024 conference, **child support award grant**: 605 €
2023	**Marie Sklodowska-Curie Actions - Postdoctoral Fellowship** (MSCA-2022-PF 101103530): 181 152.96 € (2 years); score: 98.6%
2022	**Postdoctoral fellowship *Beatriu de Pinós*** (2021 BP 00068): 144 300 € (3 years); score: 87.3%
2022/05/10	International Society Of Protistologists (ISOP) **grant** to attend Protistology Nordics: 1200 €
2017/05/15	Young Research Association (AJC, Biological Station of Roscoff) **grant** to attend InterRad XV, France: 500 €
2016 - 2019	*Agence Nationale de la Recherche* (ANR) and the Brittany Region Grants, France, to carry out doctoral studies (PhD)
2008 - 2013	Scholarship to study BSc from the National Ministry of Education, Spain

Awards and honors

2024	*Qualification* - Section 67 (*Biologie des populations et écologie*): Qualified to become a university teacher by the Minister of Higher Education and Science, France
2020/01/22	**Best oral presentation award** in Young Researchers Day (JJC) Biological Station of Roscoff, France
2019	The Micropalaeontological Society: Microfossil Image Competition & Calendar 2019
2018/12/07	**3rd best oral presentation award** in Young Researchers Day (JJC) Biological Station of Roscoff, France
2017/10/24	**Best oral presentation for Young Scientists** in InterRad XV, Japan

Skills

Languages
Spanish: Native
English: Working knowledge
French: Conversational
Portuguese & German: Basics

Programming
R & python: Advanced
Bash, UNIX/LINUX: Intermediate
HTML & CSS: Basics

Softwares
LibreOffice: Extensive experience
Inkscape: Advanced
FIJI & GIMP: User-level

Expeditions & Fieldworks

2018/10/21 - 11/21	**TAN1810** expedition, 5 weeks, *Tangaroa* Ocean Research Vessel, New Zealand
2017/09/11 - 09/24	**Bio-MOOSE** expedition, 2 weeks, *L'atalante* Ocean Research Vessel, France
2013/04/28 - 05/04	**BioCant 3** expedition, 1 week, *Sarmiento de Gamboa* Ocean Research Vessel, Spain

Teaching

2024/12/12	**Invited speaker**: 1.5 hours: Phylogenetic analyses and tree thinking, Biological Station of Roscoff, France. "Data analysis meetings"
2022/04/06	**Lecturer**: 2.5 hours: Diversity of single-celled marine eukaryotes, Uppsala University, Sweden. Bachelor students year 1
2021/11/29 - 12/03	**Invited teacher assistant**: 30 hours. Novel methods for exploring molecular microbial diversity: long-read sequencing. Masters "*Océanographie et Environnements Marins*" (OEM), Biological Station of Roscoff, France
2021/04/15	**Lecturer**: 1.5 hours: Diversity of single-celled marine eukaryotes, Uppsala University, Sweden. Bachelor students year 1
2021/01/19	**Organiser and lecturer**: 8 hours: An introduction to Sequence Similarity Networks, Uppsala University, Sweden. Systematic Biology
2018/04/12 - 04/13	**Teacher assistant**: 12 hours. Single-cell molecular approaches for studying mixotrophs, Biological Station of Roscoff, France. MixITiN MSCA-ITN
2017/11/20 - 11/24	**Teacher assistant**: 30 hours. Marine MICROorganisms: MOLecular approaches to their taxonomic and functional diversity (MICROMOL). Masters "*Océanographie et Environnements Marins*" (OEM) and "Marine Biodiversity and Conservation" (EMBC), Biological Station of Roscoff, France

Supervision

2024/04 - 2024/09	Natalia Morales, Master thesis: Integrating DNA sequencing and oceanographic simulations to trace the origins of planktonic communities in the Caribbean Sea. Master: "*Oceanografía y Gestión del Medio Marino*", Barcelona University, Spain PDF accessible via digital.CSIC: http://hdl.handle.net/10261/394488
2018/01/08 - 06/18	Elsa Gadoin, Master thesis: Molecular diversity of microalgae in symbiosis with Nassellaria and Spumellaria (Radiolaria). Master: "*Océanographie et Environnements Marins*", Sorbonne University, France

Jury member

2025/09 - present	Pablo Aragonés Suárez, Thesis: The role of phenotypic plasticity in the evolution of metazoan multicellularity. Institute of Evolutionary Biology (CSIC-UPF, Spain). Supervisors: Elena Casacuberta and Iñaki Ruiz-Trillo. **Thesis Advisory Committee**
2022/03/31	Thomas M. Huber, Master thesis: Little big data - extending plastid genome databases using marine planktonic metagenomes. Master: Erasmus Mundus Master Programme in Evolutionary Biology, Uppsala University, Sweden. **Opponent**

Scientific services

Reviewer of BMC Biology, Deep-Sea Research, ISME, Journal of Eukaryotic Microbiology, Journal of Plankton Research, MBMG, Molecular Ecology Resources, PNAS and Protist since 2020, and *recommender* in PCI microbiol.

Expert curator of PR2, Radiolaria: pr2-database.org/team/

Software contributor in GitHub: github.com/MiguelMSandin

Outreach

Lead coordinator of The Life of Retaria Seminar Series: thelifeofretaria.github.io/

2025/05/22	*Entrevista con un biólogo*: 6-8 years old students, CRA Les Mariñes, Asturies, Spain
2021/10/01 - 10/11	*Fête de la Science*: interactive poster: Rhizaria (Ocean Touch), Brest, France
2020/02/13	Introducing biodiversity and DNA to high-school students, Morlaix, France
2019/10/06	*Fête de la Science*: Introducing plankton to the public, Morlaix, France
2017/10/15	*Fête de la Science*: Introducing plankton to the public, Roscoff, France

Expert production

A total of 16 Silhouettes of Radiolaria available to the public domain for further presentations, manuscripts and outreach purposes: available at phylopic.org/

A tool for the high throughput design of specific oligonucleotides: github.com/MiguelMSandin/oligoN-design

An introduction to Sequence Similarity Networks: github.com/MiguelMSandin/SSNetworks

A practical guide to phylogenetic analyses: miguelmsandin.github.io/phylogeniesQuickStart

A set of scripts and functions to deal with common DNA fasta files and phylogenetic trees through the command line:github.com/MiguelMSandin/random

A tool to estimate the entropy of a multiple sequence alignment (published in Sandin et al., 2021) github.com/MiguelMSandin/DNA-alignment-entropy

A wet-laboratory protocol for single-cell rDNA amplification of Nassellaria (doi: 10.17504/protocols.io.t5req56) and Spumellaria (doi: 10.17504/protocols.io.xwvfpe6)

A protocol for cleaning and staining of radiolaria skeletons to be imaged in a confocal microscope (doi: 10.17504/protocols.io.t5qeq5w)

A protocol for scanning electron microscopy imaging of opaline silica single-cell skeletons (Polycystine Radiolaria: doi: 10.17504/protocols.io.ug9etz6)

Conferences

2026/04/21 Protistology Open 2026, Prague, Czech Republic
Talk: "*Environmental phylogenetics supports a steady diversification of crown eukaryotes since the mid Proterozoic*"
Poster: "*OligoN-design: A simple and versatile tool to design specific probes and primers from large datasets*"

2025/09/24 17th InterRad congress, Florence, Italy
Round table chair: "*Molecular biology for studying diversity, ecology and evolution*"
Talk: "*Is the gelatinous matrix of Nassellaria a strategy for coping with oligotrophy?*"

2025/06/23 16th International Congress of Protistology, ICOP-ISOP joint meeting, Seoul, South Korea
Invited talk: "*Revealing the diversity and evolution of Radiolaria beyond the stars of the ocean*"

2024/10/02 Comparative genomics of unicellular eukaryotes (CGUE): Interactions and symbioses, Sant Feliu de Guixols, Spain
Talk: "*Environmental phylogenetics reveals early diversification of crown eukaryotes in the Mesoproterozoic*"

2023/07/11 IX European Congress of Protistology, ECOP-ISOP joint meeting, Vienna, Austria
Talk: "*The early evolutionary history of microbial eukaryotes and the unknown diversity*"

2023/07/05 *Jornada de Biología Evolutiva*, Barcelona, Spain
Talk: "*The early evolutionary history of microbial eukaryotes and the unknown diversity*"

2023/01/20 International Society for Evolutionary Protistology, Virtual Meeting
Talk: "*The eco-evolution of microbial eukaryotes: insights from molecular environmental surveys*"

2022/09/13 16th InterRad congress, Ljubljana, Slovenia
Talk: "*Radiolaria diversity, ecology and evolution: an integrative approach*" (Presented by F. Not)

2022/05/10 Protistology Nordics, Natural History Museum of Oslo, Norway
Talk: "*Biotic interactions driving early eukaryotic evolution*"

2020/01/22 Young Researchers Day, Biological Station of Roscoff, France
Talk: "*Diversity and evolution of Radiolaria*"
Poster: "*Intra-genomic rDNA gene variability of Nassellaria and Spumellaria (Rhizaria, Radiolaria) assessed by Sanger, MinION and Illumina sequencing*"

2019/07/30 VIII European Congress of Protistology, ECOP-ISOP Joint meeting, Rome, Italy
Talk: "*Diversity and evolution of Radiolaria*"

2018/12/07 Young Researchers Day, Biological Station of Roscoff, France
Talk & Poster: "*Time Calibrated Morpho-molecular Classification of Nassellaria (Radiolaria)*"

2017/10/24 15th InterRad congress, Niigata, Japan
Talk: "*Time Calibrated Morpho-molecular Classification of Nassellaria (Radiolaria)*"

2016/12/08 Young Researchers Day, Biological Station of Roscoff, France
Flash-Talk: "*Diversity and Evolution of Nassellaria and Spumellaria (Radiolaria)*"

Seminars

2025/07/16 The Life of Retaria Seminar Series, Online Meeting
Talk: "*Calibrating the Retaria molecular clock*"

2025/01/14 MicrobeHub Seminar, Institute of Evolutionary Biology, Barcelona, Spain
Talk: "*Revealing the diversity and evolution of Radiolaria beyond the stars of the ocean*"

2024/05/10 The Life of Retaria Seminar Series, Online Meeting
Kick-off presentation, on behalf of the organising committee

2024/02/22 Ongoing and future research, Institute of Evolutionary Biology, Barcelona, Spain
Flash-talk: "*The hidden majority: Exploring environmental DNA to better understand eukaryotic diversity and evolution*"

2023/09/04 MicrobeHub Seminar, Institute of Evolutionary Biology, Barcelona, Spain
Talk: "*Gelatinous matrix, an original strategy to cope oligotrophy in Nassellaria (Radiolaria)*"

2023/06/19 MicrobeHub Seminar, Institute of Evolutionary Biology, Barcelona, Spain
Talk: "*The hidden majority: Exploring environmental DNA to better understand eukaryotic diversity and evolution*"

2022/06/22 Systematic Biology Seminar Series, Uppsala University, Sweden
Talk: "*Biotic interactions driving early eukaryotic evolution*"

2021/10/21 Systematic Biology Seminar Series, Uppsala University, Sweden
Talk: "*Eukaryotic environmental molecular diversity and its eco-evolutionary history*"

Publications

Total citations: 553 H-index: 12

Pre-prints and under review

Sandin MM, Cohen P, Morlon H, Burki F (**2025**) Environmental phylogenetics supports a steady diversification of crown eukaryotes starting from the mid Proterozoic. *bioRxiv* 2025.12.12.693929; doi: 10.64898/2025.12.12.693929. Under minor revisions in *PNAS*

PhD thesis

Sandin MM. Diversity and Evolution of Nassellaria and Spumellaria (Radiolaria). Protistology. Sorbonne Université, November, 2019. **PhD Thesis**. HAL Id: tel-03137926

Peer reviewed publications

Sandin MM, Walde M, Henry N, Forn I, Simon N, Berney C, Massana R, Richter DJ (**2026**) OligoN-design: A simple and versatile tool to design specific probes and primers from large heterogeneous datasets. *In press* in *Mol. Ecol. Resour*. Pre-print available at: 10.1101/2025.11.04.685038

Lescot M, Lezzoche N, Laux L, Romac S, Guilloux L, Chevillon E, Bodson C, Desnos C, Elineau A, Jalabert L, Llopis Monferrer N, **Sandin MM**, Vannier T, Vernette C, Villar E, Lombard F, Carlotti F, Irisson J-O, Guidi L, Bosse A, Testor P, Coppola L, Not F (**2026**) A holistic perspective on planktonic communities across the Northwestern Mediterranean Sea. *Front. Mar. Sci*. 13:1755855. doi: 10.3389/fmars.2026.1755855

Trubovitz S, **Sandin MM**, Caron DA (**2025**) Genetic variability in microbial eukaryotes reframes marine biodiversity assessment in the age of amplicon sequencing. *PLoS One*. 20(6): e032605. doi: 10.1371/journal.pone.0326053

Sandin MM, Renaudie J, Suzuki N, Not F (**2025**) Diversity and evolution of Radiolaria: Beyond the stars of the ocean. *Current Biology.* 35, 2524–2538. doi: 10.1016/j.cub.2025.04.032

Llopis-Monferrer N, Romac S, Laget M, Nakamura Y, Biard T, **Sandin MM** (**2025**) Gelatinous matrix, an original strategy to cope with oligotrophy in Nassellaria (Radiolaria). *Environ. Microbiol*. 27 (5), e70098. doi: 10.1111/1462-2920.70098

Llopis-Monferrer N, Biard T, **Sandin MM**, Lombard F, Picheral M, Elineau A, Guidi L, Leynaert A, Tréguer PJ, Not F (**2022**) Siliceous Rhizaria abundances and diversity in the Mediterranean Sea assessed by combined imaging and metabarcoding approaches. *Front. Mar. Sci.* 9:895995. doi: 10.3389/fmars.2022.895995

Sandin MM, Romac S, Not F (**2022**) Intra-genomic rDNA gene variability of Nassellaria and Spumellaria (Rhizaria, Radiolaria) assessed by Sanger, MinION and Illumina sequencing. *Environ Microbiol*. 24(7):2979-2993. doi: 10.1111/1462-2920.16081

Xiao L, Johansson S, Rughöft S, Burki F, **Sandin MM**, Tenje M, Behrendt L (**2022**) Photophysiological response of Symbiodiniaceae single cells to temperature stress. *ISME J* doi: 10.1038/s41396-022-01243-6

Burki F, **Sandin MM**, Mahwash J (**2021**) Diversity and ecology of protists revealed by metabarcoding. *Curr. Biol*. 31-19: R1267-R1280. doi: 10.1016/j.cub.2021.07.066

Sandin MM, Biard T, Romac S, O'Dogherty L, Suzuki N, Not F (**2021**) A Morpho-molecular Perspective on the Diversity and Evolution of Spumellaria (Radiolaria). *Protist*. 172, 125806. doi: 10.1016/j.protis.2021.125806

Llopis-Monferrer N, Boltovskoy D, Tréguer P, **Sandin MM**, Not F, Leynaert A (**2020**) Estimating biogenic silica production of Rhizaria in the global ocean. *Global Biogeochemical Cycles*, 34, e2019GB006286. doi: org/10.1029/2019GB006286

Nakamura Y, **Sandin MM**, Suzuki N, Somiya R, Tuji A, Not F (**2020**) Phylogenetic revision of the order Entactinaria—Paleozoic relict Radiolaria (Rhizaria, SAR). *Protist* 171. Doi: 10.1016/j.protis.2019.125712

Qui-minet ZN, Coudret J, Davoult D, Grall J, Cariou T, **Sandin MM**, Martin S (**2019**) Combined effects of global climate change and nutrient enrichment on the physiology of three temperate maerl species. *Ecol*. 00, 1–21. doi: 10.1002/ece3.5802

Sandin MM, Pillet L, Biard T, Poirier C, Bigeard E, Romac S, Suzuki N, Not F (**2019**) Time Calibrated Morpho-molecular Classification of Nassellaria (Radiolaria). *Protist* 170, 187–208. doi: 10.1016/j.protis.2019.02.002 (**Journal cover**)

Villar E, Dani V, Bigeard E, Linhart T, **Sandin MM**, Bachy C, Six C, Lombard F, Sabourault C, Not F (**2018**) Symbiont Chloroplasts Remain Active During Bleaching-Like Response Induced by Thermal Stress in Collozoum pelagicum (Collodaria, Retaria). *Front. Mar. Sci*. 5, 1–11. doi:10.3389/fmars.2018.00387

Fernandez S, **Sandin MM**, Beaulieu PG, Clusa L, Martinez JL, Ardura A, García-Vázquez E (**2018**) Environmental DNA for freshwater fish monitoring: insights for conservation within a protected area. *PeerJ* 6 e4486. doi: 10.7717/peerj.4486X

Sandin MM, Fernández C (**2016**) Changes in the structure and dynamics of marine assemblages dominated by Bifurcaria bifurcata and Cystoseira species over three decades (1977-2007). *Estuar. Coast. Shelf Sci*. 175, 46–56. doi: org/10.1016/j.ecss.2016.03.015

Other publications

Sandin MM, Renaudie J, Suzuki N, Not F (**2025**) Response to "Radiolarian evolution: Analytical challenges in estimating the diversity and origin of Nature's stars". *EcoEvoRxiv* doi: 10.32942/X2V359